

pygarble on published human-generated gibberish: local heuristics and calibration transfer

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Abstract

Inexpensive text filters can identify unusual character or word patterns, but their decisions depend on the input domain and operating threshold. We evaluate pygarble profiles and lightweight character models using published meaningful/gibberish classifications without new annotation. The primary analysis uses one 400-character prefix from each of 38 released human-generated gibberish documents and 100 blocks from each of four English comparison documents. Default English and extended profiles detect 16 and 19 positive documents, respectively. A single dictionary strategy detects all 38 with no false flags in the 400 selected control blocks. However, source-held-out calibration reveals substantial transfer failures: a 1% validation false-positive budget can yield much higher rates on another meaningful source. These results describe a small curated collection, not general accuracy or semantic understanding. We provide pinned retrieval, predictions, descriptive uncertainty, CPU measurements and a reproducible evaluation harness to support further scrutiny.

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Introduction

Gibberish screening is often an early step in text processing: a program flags suspicious material before a more expensive analysis or a decision to retain it. This can be useful when inputs contain keyboard noise, corrupted characters or unfamiliar strings. The decision is nevertheless task-specific. Historical spelling, names and specialist vocabulary can be meaningful while looking unusual to a modern English dictionary. Conversely, invented text can imitate the statistical properties of language without carrying its meaning.

pygarble supplies deterministic strategies, configurable ensembles and inspectable scores for local screening (Rao 2026). Its base installation has no runtime dependencies. A separate module handles selected PII, secret and profanity patterns; those capabilities are outside this evaluation. The present study asks a narrower question: how do existing local strategies respond to a published collection of human-generated gibberish and English comparison text, and how stable are thresholds across the available sources?

The contribution is an external, reproducible assessment of configurations and a small calibration-transfer study. It is not a new detection algorithm or an estimate of performance on arbitrary user input. The default ensemble is not assumed to outperform its individual strategies. We retain unfavorable results and distinguish the collection’s labels from a new determination of whether particular text is meaningful.

Data and evaluation design

Published classifications

Gaskell and Bown collected handwritten invented-language samples for a study of statistical resemblance to the Voynich manuscript (Gaskell and Bown 2022). Their reported collection involved 42 volunteers and additional author samples. We use the released GitHub archive at revision `d076a7d081f35098fa405928239595afd2e75927`,

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which contains 38 gibberish transcript files. We include all 38 and do not infer their individual cohort membership. The study evaluates the available release, not the original study’s complete collection or its Voynich conclusions.

We select the four comparison files explicitly identified as English: the King James Version New Testament (KJV), a modern NET Bible text, a historical English Secreta Alberti transcription, and a Wikipedia-derived Voynich article. Their supplied meaningful/gibberish classifications are inherited without new annotation or human label audit. This does not establish that every comparison passage is clean modern prose. Non-English comparison files and Voynichese are excluded without classifying them as gibberish.

Source rights differ. The gibberish data carry the authors’ modified MIT notice and citation requirement; meaningful comparison texts retain their original terms. Our repository publishes retrieval instructions, hashes, offsets and measurements rather than redistributing the comparison texts. No new participants were recruited and no participant identities were inferred.

Inputs and dependence

Whitespace is collapsed to single spaces while case, accents and punctuation are retained. The primary input is exactly 400 Unicode characters. We take one prefix per gibberish document and up to 100 evenly spaced, nonoverlapping blocks per comparison document. All four controls provide 100 primary blocks. This controls input length without producing multiple nominally independent positive examples from each transcript. A prefix can end within a word; the same fixed-length slicing rule applies to both classes.

Predeclared sensitivity views use 100 and 800 characters, with documents too short for a complete block omitted. The 800-character view has 36 positive documents and 362 control blocks. Whole positive documents are an additional endpoint. These overlapping views are dependent and are not independent replications. Source/class confounding remains: gibberish and comparison prose were produced under different conditions.

An automated check found no exact duplicate primary blocks and no primary block with at least half its five-word shingles covered by the repository’s existing JSON regression material. Across separate control families, maximum five-word-shingle containment was below 0.1%. The Bible translations share content and are kept in one family despite their different wording. These checks cannot establish independence from every dictionary or historical resource used in detector development.

Methods

Fixed configurations

We evaluate pygarble’s **english**, **english_extended** and **legacy** profiles with their default threshold of 0.5, **any** voting and one thread. The English profile combines its legacy language signals with encoding/control-character and keyboard signals; the extended profile adds pattern, local-anomaly and repetition checks. The legacy profile provides a fixed strategy-group comparison. Single **word_lookup** and **entropy_based** strategies use their default settings. An always-keep policy establishes the trivial no-removal reference. These package strategies share implementation/resources and are not presented as independent software systems.

For calibration experiments, two additional baselines fit conditional character models locally. The bigram baseline adapts Renaud’s approach (Renaud, n.d.): retain lower-case ASCII letters and spaces, add ten pseudocounts per character transition, and average log transition likelihoods. We reverse the sign so that larger scores indicate less resemblance to training prose. The trigram extension uses two-character contexts and add-one smoothing. Neither uses pretrained weights or an LLM. Training data and threshold selection differ from the upstream program; these are documented adaptations, not claims to reproduce its original accuracy.

Source-held-out calibration

The controls form three families: Bible (both translations), Secreta and Wiki. The fixed rotations assign distinct families to testing, training and validation:

Test family	Character-model training	Threshold validation
Bible	Wiki	Secreta

Test family	Character-model training	Threshold validation
Secreta	Bible	Wiki
Wiki	Secreta	Bible

Character models use the first 100,000 normalized characters from each training document, with no transitions across documents. Package resources remain unchanged. No gibberish transcript is used to fit any model or select a threshold. On meaningful validation blocks, select the lowest candidate threshold producing at most 1% false positives. Candidates are zero and the next representable floating-point number above each distinct applicable score; decisions use $\geq$, preserving ties. This policy maximizes flags subject to the empirical constraint without optimizing on positive test data.

The held-out meaningful family and the same 38 positive prefixes are then scored. Those positives recur across folds; they are not pooled as 114 independent observations. The available three control families are too few for a broad supervised-classifier comparison, so a trained linear classifier from the initial planning document was excluded before scoring. No model or threshold was selected after inspecting final outcomes.

Metrics and execution

We report positive-document recall, false-positive counts and rates within each control document, and macro-average control-document FPR where useful. The collection’s artificial class prevalence makes overall accuracy, precision and F1 unsuitable as headline deployment measures. Inapplicable positive results count as misses; coverage is also recorded.

Recall intervals use the 95% Wilson construction conditional on independent documents. A paired document bootstrap with 2,000 replicates and seed 20260926 provides descriptive recall differences against default English. Unknown participant dependence limits these intervals. We do not calculate passage-binomial FPR confidence intervals: hundreds of blocks still come from four documents. No multiplicity-adjusted significance claim is made.

The protocol and harness were committed as 22f30d8 before detector predictions. Scripts use the standard library and the unchanged local package. A fresh virtual environment reproduced the deterministic prediction and summary artifacts byte for byte. Scoring requires no external inference; source downloads occur during preparation only.

Results

Default decisions

Method	Detected	Recall	KJV FPR	NET FPR	Secreta FPR	Wiki FPR
Always keep	0/38	0.0%	0%	0%	0%	0%
English	16/38	42.1%	0%	1%	3%	1%
Extended	19/38	50.0%	1%	1%	3%	1%
Legacy	14/38	36.8%	0%	0%	0%	1%
Word lookup	38/38	100.0%	0%	0%	0%	0%
Entropy	0/38	0.0%	0%	0%	0%	0%

The dictionary strategy detected all released positive prefixes while the default English ensemble missed 22 of 38. This is evidence that the ensemble is not uniformly stronger than a single lexical signal on this collection. The dictionary recall interval is approximately 90.8–100% under the stated independence assumption. Its zero observed false positives across four control documents does not establish a zero population FPR.

The extended profile’s recall increase over English is 7.9 percentage points; the paired bootstrap interval is 0.0–15.8 points. Coverage was 100% for these views. The entropy strategy made no positive decisions at its default cutoff. That result describes this implementation and threshold, not a claim that entropy cannot contribute to another method.

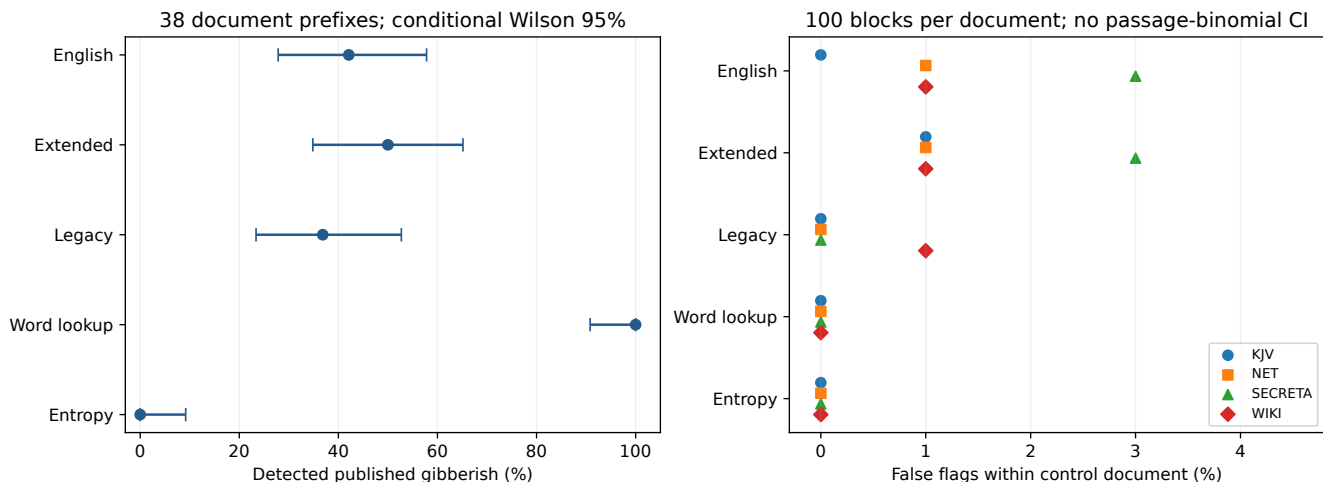


Figure 1: Default 400-character results. Recall intervals are conditional on document independence. Control false-positive rates are descriptive within-source measurements.

Length sensitivity

Method	100 chars	400 chars	800 chars	Whole positive
Always keep	0/38	0/38	0/36	0/38
English	12/38	16/38	14/36	17/38
Extended	15/38	19/38	21/36	26/38
Legacy	12/38	14/38	12/36	14/38
Word lookup	38/38	38/38	36/36	38/38
Entropy	0/38	0/38	0/36	0/38

The extended profile detects more complete transcripts than 400-character prefixes, consistent with different opportunities to trigger its strategies. Because the whole-document endpoint has no matched control length, it cannot establish improved overall discrimination. Length-view denominators and control rates are available in the machine-readable results.

Calibration transfer

Method	Bible recall / FPR	Secreta recall / FPR	Wiki recall / FPR
English	21.1% / 0.5%	44.7% / 3.0%	60.5% / 13.0%
Extended	42.1% / 1.0%	52.6% / 3.0%	57.9% / 7.0%
Legacy	68.4% / 4.0%	42.1% / 0.0%	60.5% / 13.0%
Word lookup	100.0% / 0.0%	100.0% / 99.0%	100.0% / 3.0%
Entropy	15.8% / 1.0%	50.0% / 29.0%	21.1% / 0.0%
Character bigram	100.0% / 4.0%	100.0% / 1.0%	100.0% / 42.0%
Character trigram	100.0% / 0.0%	100.0% / 2.0%	100.0% / 44.0%

The 1% validation constraint did not generally transfer. For example, calibrating word lookup on Wiki before testing Secreta produced 99 false flags in 100 Secreta blocks, compared with none at the original default threshold. Both character models detected all 38 positives in every fold, but their Wiki false-positive rates were 42% and 44%. Their training family was Secreta and their validation family was Bible in that fold.

These differences show why low false-positive behavior on one source is insufficient to promise the same behavior on another. They do not establish which threshold is optimal in a future application. Outcomes can reflect both source domain and the fixed assignment of training and validation families. Exhaustive rotations or alternative assignments would be additional exploratory analyses and are not implied by this study.

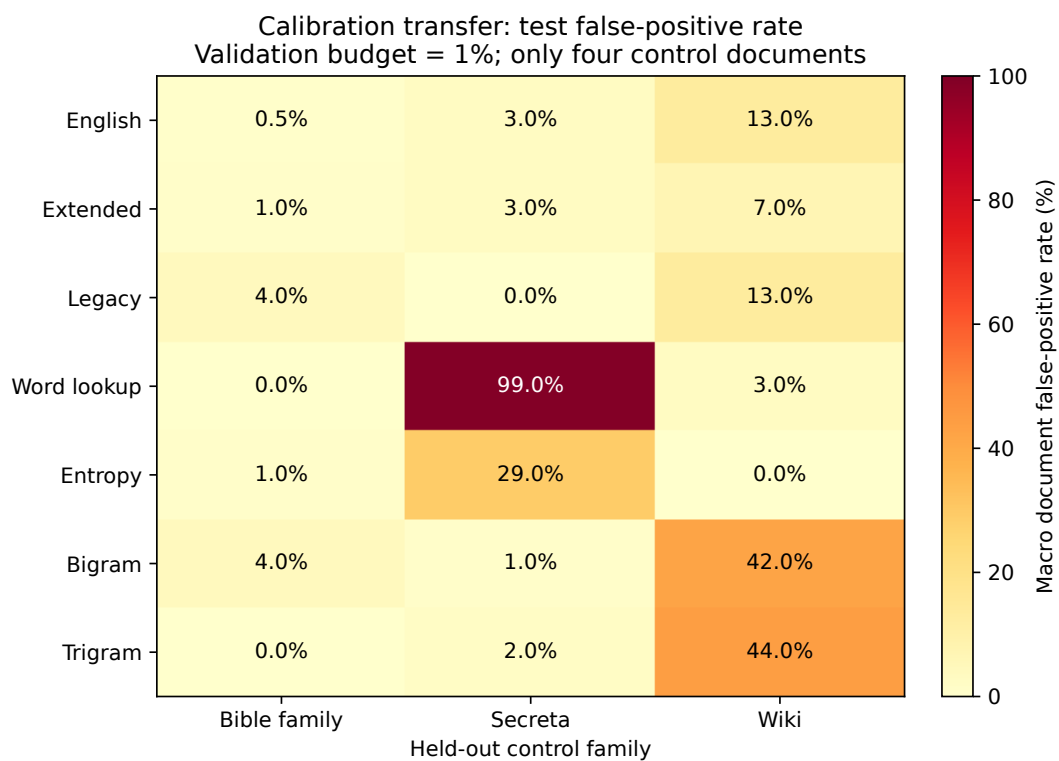


Figure 2: Test false-positive rates after calibration on a different family. Every validation rate meets the empirical 1% budget; test rates can be much higher.

CPU and memory measurements

Measurements used an Apple M2 with 8 GiB RAM, macOS and Python 3.12.2. Each method ran in a fresh process on 76 fixed 400-character inputs: 38 positive prefixes and 38 deterministically selected controls. After an unmeasured warm pass, five timed passes measured the complete score-and-applicability path. This differs from the library’s potentially faster short-circuit `predict` path. Package import, construction/training and first-call measurements are stored separately. Peak RSS was measured in separate worker processes and includes Python, imports and resident corpus data.

Method	Median ms	p95 ms	Process peak MiB
English	0.571	0.691	52.6
Extended	1.075	1.274	57.9
Legacy	0.408	0.778	58.4
Word lookup	0.038	0.109	55.0
Entropy	0.151	0.177	50.0
Character bigram	0.147	0.220	49.9
Character trigram	0.166	0.180	50.2

Character-model timings use the Secreta-trained fit. Measurements come from one machine and a fixed input order; the trivial always-keep cost approaches timer overhead. They are not production latency guarantees or cloud-cost estimates. No paid data, inference or cloud-compute service was procured.

Discussion and limitations

The most direct finding is modest: lexical unfamiliarity separates this published invented-language collection from the selected English controls more strongly than several existing default profiles. That is compatible with a collection built from intentionally invented words. It does not show that the dictionary method recognizes semantic nonsense made entirely of ordinary words, and it does not justify replacing a production default on this evidence alone.

The study also provides a concrete counterexample to interpreting an empirical calibration budget as a general guarantee. A threshold learned on modern technical text can reject historical meaningful spelling. Even with all positives detected, a method can be unsuitable for automatic removal because it flags substantial meaningful material. Reusing the supplied classifications made the study inexpensive but leaves label suitability unverified; no new human audit was performed.

Four control documents and 38 released positive documents cannot represent the diversity of real inputs. Limited source diversity, prefix selection, missing participant metadata, untested languages and possible overlap with pre-existing lexical resources all constrain inference. This is a secondary analysis of an existing collection, not independent confirmation of its labels. The absence of a domain-matched deployment corpus also prevents claims of downstream research-quality improvement.

For software users, the practical implication is to expose configuration and source-specific errors rather than equating a composite detector with a universal answer. The reproducible harness makes those choices inspectable. A larger independently sourced evaluation and actual adopter feedback would be needed for stronger claims about broad research utility. No journal acceptance or indexing outcome follows from the present results.

Availability and author review

Code, pinned source retrieval, data notices, predictions, figures and reproduction instructions are in [paper/study/](#) of [the pygarble repository](#). The raw comparison corpus is not republished. Exact script/library hashes are recorded in [results/code-manifest.json](#).

The author reports using OpenAI Codex and Anthropic Claude in the wider project. Codex (GPT-6 in this session) assisted with study design, research code, automated execution, analysis and manuscript drafting. No LLM was used for detector inference or creation of evaluation labels. Claude’s precise scope and model versions remain to be confirmed. Human review of this draft and responsibility for its scientific decisions have not yet been asserted.

Funding and conflicts of interest: author declarations pending. This draft must be reviewed and these declarations completed before any submission. A prospective JOSS article would describe the software contribution and reference this evaluation, rather than duplicate a results-focused article.

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